

Project Olympus · Unit 0: Character Creation

Teacher's Edition with answer keys, hints, and solutions · The Language of Everything · David Balyeat, with Claude

One week. Set up the tools, learn the rules, win early. The math here is **estimation**: getting to a good-enough answer fast, and knowing when an answer smells wrong. Physicists call these Fermi problems. You'll call them easy by Friday.

Week 1 · Day 1: Welcome to Olympus

The rules of the game, and a trick that is secretly the whole course

Warm-up

No calculator. Is 19×21 closer to **400** or **4,000**?

Answer: 400. 19×21 is a hair under $20 \times 20 = 400$. (It's exactly 399.) You just did an estimate. That's today's whole idea.

The Story

Why this exists

You tested past high school. Your school math is doing long division. So we're skipping the part where math is a pile of rules to memorize and going straight to the part where math is a **language**: the one that describes how a free kick curves, how money grows, why a breakfall works, and how far away the Moon is.

The catch: a language only works if you actually speak it. So you'll write on paper, draw pictures, build spreadsheets, run experiments, and explain things out loud. The screen checks your answers. The notebook is where the learning happens.

THE STORY · RICHARD FEYNMAN AND THE BIRD

When Feynman was a kid, his dad took him on walks in the woods. Another kid once pointed at a bird and said, "That's a brown-throated thrush. Your father doesn't teach you anything!" Feynman's dad had actually told him: "You can know the name of that bird in every language in the world, and when you're finished you'll know absolutely nothing about the bird. Now let's *look* at the bird."

That's the deal here. Names of things (slope, derivative, logarithm) are cheap. We'll define every one in plain English. But the point is always to look at the bird.

The rules of the game

- **XP** for every problem. Full XP first try, no hints. Hints cost 20% each, wrong tries cost 10%. Retries always allowed.

- **Ranks:** Mortal → Hero → Demigod → Titan → Olympian.
- **Badges** from Labors (the boss fight at the end of each unit) and from real-world stuff: teaching, performing, catching the AI, building.
- **Quests:** treasure hunts around the house. Real lock, real box.
- **Beat Dad Friday:** one problem, same clock.
- **The Codex:** your own dictionary. Writing your own definition of a term earns XP.

DECODER

Estimate

A smart guess you can defend. Not a random guess: a quick calculation with rounded numbers that gets you close enough to make a decision.

" 19×21 is about 400" is an estimate. " 19×21 is about a million" is a guess by someone who hasn't looked at the bird.

DECODER

Order of magnitude

Which power of ten a number is closest to: 1, 10, 100, 1,000, 10,000... Two numbers are "in the same order of magnitude" if one is less than about 10 times the other.

A soccer match (90 min) and a jiu-jitsu class (60 min) are the same order of magnitude. A match and a school year are not.

The three rules of tools

Estimate → Calculate → Check. Before you touch a calculator, guess the answer to one digit. Then calculate. Then ask: is the calculator's answer near my guess? If not, one of you is wrong, and it's usually the typing. This is the habit that separates people who use tools from people who get used by them.

MATHEMAGIC · "I CAN READ YOUR MIND"

Try this on someone tonight. Tell them:

1. Think of a number. Any number. Don't tell me.
2. Double it.
3. Add 10.
4. Divide by 2.
5. Subtract the number you started with.

Then say: "Your answer is **5**." It always is. Every time. For any number.

Why? Call their secret number **n**. Watch what happens:

Step	What they have
Think of a number	n
Double it	2n

Add 10	$2n + 10$
Divide by 2	$n + 5$
Subtract the original	5

The n cancels out. It was never going to matter. **That is algebra.** A letter stands for "a number I don't know," you follow the rules, and the truth falls out. Justin Willman would sell this with a lot more eyebrow. But the engine underneath is the same as the engine under every unit in this course.

DECODER

Variable

A letter that stands for a number you don't know yet, or a number that could be anything. In the trick, n was the number the other person was thinking of.

In " $2n + 10$ ", n is the variable. The 2 and the 10 are just numbers.

ON PAPER

In your notebook: write the title **The Codex** on the first page. Under it, write your own definitions of *estimate*, *order of magnitude*, and *variable*. Then write out the mind-reading trick with the n -table. You'll perform it this week.

Practice

1. Without a calculator: 48×52 is closest to which?

- A. 250
- B. 2,500
- C. 25,000
- D. 250,000

Answer: 2,500 (B)

Hints: (1) Round both numbers to the nearest 10. (2) $50 \times 50 = ?$ (3) Then decide which choice is nearest to that.

Solution: $50 \times 50 = 2,500$. (Exact: 2,496.) The answer choices are $10\times$ apart, so a rough estimate is all you need.

2. The number of seconds in a day is closest to which order of magnitude?

- A. 1,000
- B. 10,000
- C. 100,000
- D. 1,000,000

Answer: 100,000 (C)

Hints: (1) 60 seconds in a minute, 60 minutes in an hour, 24 hours in a day. (2) $60 \times 60 = 3,600$ seconds per hour. (3) $3,600 \times 24 \approx 3,600 \times 25 = 90,000$.

Solution: $60 \times 60 \times 24 = 86,400$ seconds. That's closest to 100,000 (10^5).

3. A soccer pitch is about 105 m long and 68 m wide. Hudson multiplies and gets an area of **71,400 m²**. Is that reasonable?

- A. Reasonable
- B. Too big by about 10×
- C. Too small by about 10×
- D. Too big by about 100×

Answer: Too big by about 10× (B)

Hints: (1) Estimate first: 100×70 . (2) $100 \times 70 = 7,000$. (3) Compare 7,000 to 71,400.

Solution: $105 \times 68 = 7,140 \text{ m}^2$. He got 71,400: ten times too big. A slipped decimal. The estimate caught it in two seconds, which is the point.

4. A new version of the trick: think of a number, **triple it, add 12, divide by 3**, then **subtract your original number**. What does everyone get?

Answer: 4

Hints: (1) Call the number n and build the table like in the story. (2) Triple: $3n$. Add 12: $3n + 12$. Divide by 3: $n + 4$. (3) Now subtract n .

Solution: $n \rightarrow 3n \rightarrow 3n + 12 \rightarrow n + 4 \rightarrow 4$. The 12 got divided by the 3, and the n vanished.

5. In the original trick (double, add 10, halve, subtract original), the magician changes "add 10" to "add 20." Now what's the answer everyone gets?

Answer: 10

Hints: (1) Follow the table with 20 instead of 10. (2) $2n + 20$, divided by 2, is $n + 10$.

Solution: $2n \rightarrow 2n + 20 \rightarrow n + 10 \rightarrow$ subtract $n \rightarrow 10$. Whatever you add gets halved. So the answer is always half of what you add.

6. Hero rank is **1,500 XP**. Suppose a session earns about **90 XP**. About how many sessions to Hero?

Answer: 17 sessions (± 1)

Hints: (1) Divide total by per-session. (2) $1,500 \div 90$. Estimate: $1,500 \div 100 = 15$, and 90 is a bit smaller than 100, so a bit more than 15.

Solution: $1,500 \div 90 = 16.7$, so about **17 sessions**. That's roughly the end of Unit 1. Good timing for a rank-up.

7. Spider-Man web-swings across Queens at roughly **25 m/s** (about 55 mph; the movies are vague, so we're estimating). About how many seconds to cover **1 kilometer**?

Answer: 40 s (± 1)

Hints: (1) 1 kilometer = 1,000 meters. (2) Time = distance $\div$ speed. (3) $1,000 \div 25$.

Solution: $1,000 \text{ m} \div 25 \text{ m/s} = \mathbf{40 \text{ s}}$. Quick check: $40 \times 25 = 1,000$. ✓ (Notice the units: meters divided by meters-per-second leaves seconds. Units are a built-in error detector; we'll lean on that all year.)

Apply: Set up your gear

Today's Apply is setup. Boring, but it's a one-time thing and it earns XP.

- Notebook.** Page 1: "The Codex." Page 2: the mind-reading trick with its n-table. Leave the first 10 pages for the Codex; start session notes on page 11.
- Profile.** You're already logged in as you. Dad has his own. Don't touch his. (He's a Mortal too.)
- Calculator.** Find the parentheses keys, the x^2 key, the $^$ or x^y key, and the $\times 10^n$ (sometimes EE or EXP) key. Type 2^{10} and get 1024.
- Claude.** Dad sets up your tutor (see Dad notes). Your first prompt: "I'm starting a math program. Give me one hint, not the answer, for this: $1,000 \div 25$." See what it does.
- Perform the trick** on one family member tonight. Tomorrow, on a second one. Willman badge on the line: perform it AND explain the n-table.

Dad notes: Create a Claude Project for Hudson (or a saved custom instruction) and paste in the tutor profile from the Parent Guide appendix. Test it yourself first: ask it to solve $3x + 2 = 17$ and confirm it gives hints, not the answer, until you've tried twice. Set the Dad PIN in this workbench (top-right chip). Tonight: let him perform the trick on you. Act amazed. Then ask "why does it work?" and don't accept "because math."

Week 1 · Day 2: Fermi's Napkin

How physicists estimate anything in 60 seconds

Warm-up

Powers of ten: $10^3 \times 10^2 = ?$ (Type the plain number.)

Answer: 100,000, which is 10^5 . Multiplying powers of ten means *adding* the exponents: $3 + 2 = 5$. That one rule does most of today's heavy lifting.

The Story

THE STORY · ENRICO FERMI AND THE CONFETTI

July 16, 1945. The first atomic bomb test, in the New Mexico desert. Enrico Fermi, one of the great physicists of the century, stood about 10 miles away holding a handful of torn-up paper. When the blast wave arrived, he dropped the scraps and watched how far they blew: about 2.5 meters. From that, in his head, he estimated the bomb's power at around 10 kilotons of TNT. The official measurement, after weeks of analysis with instruments, was about 21 kilotons.

Fermi got within a factor of two. With confetti.

(Feynman was there too. He decided the truck windshield would block the dangerous ultraviolet light and watched the explosion through it, without the dark goggles everyone else wore. He was probably the only person who saw it directly. Don't do that.)

Fermi was famous for questions like "How many piano tuners are in Chicago?" You can't look it up (well, you can now, but that's cheating), and you can't know it. But you can **reason** your way to a number that's about right. These are called **Fermi problems**, and they're the single most useful math skill for real life, because real life almost never gives you exact numbers.

The Fermi method

- 1 Break it into pieces you can guess.** Piano tuners = (people in Chicago) $\times$ (pianos per person) $\times$ (tunings per piano per year) $\div$ (tunings one tuner does per year).
- 2 Guess each piece to one significant figure** or to the nearest power of ten. 3 million people. 1 piano per 20 people. 1 tuning per year. A tuner does 4 a day $\times$ 250 days = 1,000 a year.
- 3 Multiply it out.** $3,000,000 \times 1/20 \times 1 \div 1,000 = 150$ tuners.
- 4 Check units and sanity.** 150 tuners for a big city? Sounds plausible. (Real answer: somewhere around there.)

Scoring: within 3× of the true answer is a win. Within 10× is fine. The goal is the right order of magnitude, not the right digits.

DECODER

Scientific notation

Writing a number as (a number between 1 and 10) × (a power of ten). It keeps huge and tiny numbers readable and makes multiplying them easy: multiply the front numbers, add the exponents.

$35,000 = 3.5 \times 10^4$. And $(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^7$. On a calculator, type 3.5 then the $\times 10^n$ key then 4. In Sheets or Python, write 3.5e4.

DECODER

Significant figure

The digits in a number that actually carry information. "About 300" has one significant figure. "312" has three. In a Fermi problem, one is enough, and pretending you have more is lying to yourself.

If you estimate a room at 4 m wide, writing the area as 16.4372 m² is theater. It's 16. Maybe.

WORKED EXAMPLE · HOW MANY RUBIK'S CUBES WOULD FILL YOUR BEDROOM?

Pieces: volume of the room ÷ volume of one cube.

Cube: a standard 3×3 is 5.7 cm on a side. Round to 6. Volume $\approx 6 \times 6 \times 6 \approx 200 \text{ cm}^3$ (5.7³ is 185; close enough).

Room: say 4 m × 4 m × 2.5 m = 40 m³. One cubic meter is 100 × 100 × 100 = 1,000,000 cm³. So the room is $4 \times 10^7 \text{ cm}^3$.

Divide: $4 \times 10^7 \div 2 \times 10^2 = 2 \times 10^5 = \mathbf{200,000 \text{ cubes}}$.

Sanity: a cube is roughly a coffee mug. 200,000 mugs in a bedroom? A full room of mugs stacked to the ceiling... yes, that's about right. Also: that many cubes at \$10 each is \$2 million. Ask for a smaller room.

GREEK ORIGINS · ARCHIMEDES COUNTS THE SAND

Around 250 BC, Archimedes wrote *The Sand Reckoner* to answer a question the king thought was impossible: how many grains of sand would fill the entire universe? Greek numerals topped out around 10,000, so he invented a way to write enormous numbers first (basically scientific notation, 2,000 years early), then estimated: about 10⁶³ grains. He was doing Fermi problems before Fermi's great-great-great-great-grandparents were born.

He also reportedly ran naked through Syracuse shouting "Eureka!" after solving a problem in the bath. Feynman played bongos. Great scientists are allowed to be weird.

Practice

1. How many **seconds are in a year**? Estimate it (no calculator until you've guessed), then calculate.

Answer: about 31,536,000 (anything from 10,000,000 to 100,000,000 counts)

Hints: (1) Seconds per day you already know from yesterday: about 86,400, or call it 10^5 . (2) Days in a year: 365, call it 4×10^2 . (3) $10^5 \times 4 \times 10^2 = 4 \times 10^7$. Now the exact one: $86,400 \times 365$.

Solution: $86,400 \times 365 = 31,536,000 \approx 3.15 \times 10^7$. The Fermi estimate (4×10^7) was within 30%. Physicists actually remember this as " $\pi \times 10^7$ seconds," which is a coincidence but a handy one.

2. How many times has your **heart beaten** in your life so far? (You're 13. Resting heart rate is about 70 beats per minute; you've been asleep, running, and sitting, so 70 is a fair average.)

Answer: about 480,000,000 (anything from 100,000,000 to 1,000,000,000 counts)

Hints: (1) Beats per minute $\times$ minutes per hour $\times$ hours per day $\times$ days per year $\times$ years. (2) $70 \times 60 \times 24 = 100,800$ beats per day. Call it 10^5 . (3) 10^5 per day $\times$ 365 days $\times$ 13 years.

Solution: $70 \times 60 \times 24 \times 365 \times 13 \approx 4.8 \times 10^8$, about **480 million beats**. Half a billion. Every one of them without you asking. Unit 8 is about that machine.

3. How many **kicks of a soccer ball** happen worldwide in one day? (FIFA estimates about 250 million people play soccer regularly.)

Answer: about 2,500,000,000 (anything from 100,000,000 to 100,000,000,000 counts)

Hints: (1) Pieces: (players) $\times$ (fraction who play on a given day) $\times$ (kicks per player per session). (2) Say 1 in 10 play on any given day: 25 million players today. (3) A training session or match: maybe 100 touches each.

Solution: $250,000,000 \times 1/10 \times 100 = 2.5 \times 10^9$: about **2.5 billion kicks a day**. Your assumptions may differ (1 in 5? 50 touches?) and that's fine; anything from 10^8 to 10^{11} is defensible. Write down your assumptions. That's what makes it an estimate and not a guess.

4. How many **jiujitsu sparring rounds** are happening worldwide during one of your one-hour classes? (Rough guess: about 3 million people train BJJ worldwide.)

Answer: about 150,000 (anything from 10,000 to 1,000,000 counts)

Hints: (1) Pieces: (people training) $\times$ (fraction on the mat during this particular hour) $\times$ (rounds per person per hour) $\div$ (2 people per round). (2) Fraction training right now: classes run maybe 3 hours a day out of 24, people train maybe 3 days a week out of 7, so roughly 1 in 50 are on a mat at any given hour. (3) $3,000,000 \div 50 = 60,000$ people on mats. Each does about 5 rounds an hour. Two per round.

Solution: $60,000 \times 5 \div 2 = \mathbf{150,000 \text{ rounds}}$ in that hour, give or take a lot. Somewhere on Earth, right now, someone is getting caught in a triangle. Comforting.

5. How many **Lego bricks** would a **life-size Millennium Falcon** need, if you built it solid? The Falcon is about 35 m long, 25 m wide, and 8 m tall (but it's a saucer, not a box: use about 1/3 of the box volume). A standard 2×4 brick is 3.2 cm × 1.6 cm × 1 cm.

Answer: about 400,000,000 (anything from 100,000,000 to 1,000,000,000 counts)

Hints: (1) Falcon volume: $35 \times 25 \times 8 \times (1/3)$ cubic meters. (2) That's about $2,300 \text{ m}^3$, or $2.3 \times 10^9 \text{ cm}^3$ (a cubic meter is 10^6 cm^3). (3) Brick volume: $3.2 \times 1.6 \times 1 \approx 5 \text{ cm}^3$. Divide.

Solution: $2.3 \times 10^9 \div 5 \approx 4 \text{ to } 5 \times 10^8$ **bricks**: half a billion. Friday's Labor checks this a totally different way, using the actual Lego set. If two different methods agree, you're probably right. That's a real scientist move.

6. Calculate: $(3 \times 10^4) \times (2 \times 10^3)$. Type the plain number.

Answer: 60,000,000

Hints: (1) Multiply the front numbers: 3×2 . (2) Add the exponents: $4 + 3$.

Solution: $6 \times 10^7 = 60,000,000$.

7. Your Fermi estimate for something was 4×10^8 . The true value turned out to be 5.2×10^8 . By what factor were you off? (Divide the bigger by the smaller.)

Answer: 1.3 (± 0.05)

Hints: (1) $5.2 \div 4$.

Solution: $5.2 \div 4 = 1.3$. Off by 30%. For a Fermi problem that is a bullseye.

Apply: Your own Fermi problem

Make one up about something you care about. Examples: How many Fortnite matches are played per day? How many Lego bricks exist on Earth? How many hours has Messi spent with a ball at his feet?

1. Write the question in your notebook.
2. List the pieces (3 to 5 of them) and your guess for each, with units.
3. Multiply. Write the answer in scientific notation with one significant figure.
4. Sanity check: does it feel right? Can you compare it to something you know?
5. Paste your pieces and guesses into Claude and ask: "Critique my assumptions. Don't give me a better answer; tell me which guess is weakest and why."

Dad notes: Push on the assumptions at dinner. The skill isn't the arithmetic; it's noticing which guess the whole answer hangs on. If his Fortnite estimate depends on "how many people play per day," make him defend that number.

Week 1 · Day 3: The Three Tools

Calculator, Sheets, and AI: how each one lies to you, and how to catch it

Warm-up

No calculator: $2 + 3 \times 4 = ?$

Answer: 14. Multiply before you add. If you got 20, your calculator would have agreed with you only if you typed it wrong. Today is about that.

The Story

Tool 1: the calculator

A calculator answers exactly the question you typed. Not the one you meant. The gap between those two is where every calculator error lives.

DECODER

Order of operations

The agreed order for doing a calculation: Grouping (parentheses) first, then Exponents, then Multiply and Divide (left to right), then Add and Subtract (left to right). People remember it as PEMDAS or GEMS. Calculators follow it whether you meant to or not.

$10 \div 2 \times 5 = 25$, not 1. Left to right: $10 \div 2 = 5$, then $\times 5 = 25$. If you meant $10 \div (2 \times 5)$, you need the parentheses.

Keys to know: parentheses; x^2 and $^$ (or x^y) for exponents; $\times 10^n$ (sometimes EE or EXP) for scientific notation; ANS for the last answer; and the fraction key if you have one. **Habit:** estimate the answer to one digit before you type. If the calculator disagrees by $10\times$, you mistyped.

Tool 2: Google Sheets

Sheets is a calculator that remembers and repeats. Every box is a **cell** with an address (A1, B7). Type a number and it's a number. Type **=** first and it's a **formula**: `=A1*B1` multiplies two cells. Change A1 and the answer updates by itself. Drag the little square at the corner of a cell downward and the formula copies to every row (**fill down**). That's most of what a financial analyst does all day, and you'll be able to do it by Friday.

WORKED EXAMPLE · THE RUBIK'S CUBE ESTIMATE, AS A SHEET

	A	B
1	Room length (cm)	400
2	Room width (cm)	400
3	Room height (cm)	250

4	Cube side (cm)	5.7
5	Room volume	=B1*B2*B3
6	Cube volume	=B4^3
7	Cubes that fit	=B5/B6

Now change the room size and the whole thing recalculates. That's the power: you build the *method* once and reuse it forever.

Tool 3: the AI

DECODER

Large language model (LLM)

The kind of AI you're talking to when you use Claude. It was trained on an enormous amount of text and learned to predict what words come next. It's very good at explaining, and it can do a lot of math, but it doesn't "know" your answer is right the way a calculator does. It can sound completely confident and be wrong.

Ask it "what's 17×23 ?" and it will probably say 391 (correct). Ask it something with lots of steps and it may drop a digit and never notice. That's the Bug Hunter opportunity.

The rules for using it well:

- 1 Hints, not answers.** "Give me a hint, not the answer." It will obey. If it doesn't, tell it to try again.
- 2 Show your work.** Paste your steps or a photo of your notebook page and ask "where's my mistake?" It is excellent at this.
- 3 Verify numbers.** Anything numeric it tells you, check with the calculator or Sheets. Every time. It takes ten seconds.
- 4 Catch it.** When it's wrong and you can prove it, screenshot it. That's a Bug Hunter badge.
- 5 Explain back.** The best prompt in the world: "Here's how I'd explain this to a 10-year-old: ... Is that right? What did I miss?"

THE STORY · FEYNMAN ON FOOLING YOURSELF

Feynman told a graduating class at Caltech: "The first principle is that you must not fool yourself, and you are the easiest person to fool." Every tool in this course is a way to *not* fool yourself: the estimate checks the calculator, the calculator checks the AI, the Sheet checks the calculator, and your notebook checks all of them.

ON PAPER

Codex entries today: *order of operations, cell, formula, fill down, large language model*.
Your words, not mine.

Practice

1. Estimate first, then use the calculator: $20 - 8 \div 2 \times 3$

Answer: 8

Hints: (1) Divide and multiply before subtracting, left to right. (2) $8 \div 2 = 4$, then $4 \times 3 = 12$.

Solution: $20 - 12 = 8$. If you got 18, you subtracted first: $(20 - 8) \div 2 \times 3 = 18$. The calculator only does that if you type the parentheses.

2. $2 + 3^2 \times (4 - 1)$

Answer: 29

Hints: (1) Parentheses first: $(4 - 1) = 3$. (2) Exponent next: $3^2 = 9$. (3) $9 \times 3 = 27$, then add 2.

Solution: $2 + 9 \times 3 = 2 + 27 = 29$.

3. In a Sheet, cell A1 holds **12** and B1 holds **5**. What does `=A1*B1+2` show?

Answer: 62

Hints: (1) Replace the cell names with their numbers. (2) $12 \times 5 + 2$, with order of operations.

Solution: $12 \times 5 = 60$, $+ 2 = 62$. (If you wanted 12×7 , you'd write `=A1*(B1+2)`.)

4. A1 = 100, A2 = 200, A3 = 300. What does `=AVERAGE(A1:A3)` show?

Answer: 200

Hints: (1) A1:A3 means "A1 through A3", all three cells. (2) Average = sum $\div$ count.

Solution: $(100 + 200 + 300) \div 3 = 200$.

5. What does `=2^10` show?

Answer: 1,024

Hints: (1) ^ means "to the power of." (2) $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

Solution: **1,024**. Memorize this one; it's why a kilobyte is 1,024 bytes and why Unit 7's binary card trick works.

6. An AI tells you: "Lionel Messi's top speed at the 2026 World Cup was 40 km/h." Sofascore's recorded figure was 30.9 km/h. What's the right move?

- A. Trust the AI; it's usually right
- B. Check the source, record the AI's mistake, and claim the Bug Hunter badge
- C. Average the two numbers
- D. Ask the AI again until it agrees with you

Answer: Check the source, record the AI's mistake, and claim the Bug Hunter badge (B)

Hints: (1) Which option involves actually checking?

Solution: Check the source and log the miss. (40 km/h would make Messi faster than any human ever recorded; Usain Bolt peaked around 44 km/h. Your sanity check should have flagged it before you even looked it up.)

7. Use the $\times 10^n$ key: $(4 \times 10^6) \div (8 \times 10^2)$. Type the plain number.

Answer: 5,000

Hints: (1) Front numbers: $4 \div 8 = 0.5$. (2) Exponents: $6 - 2 = 4$. So 0.5×10^4 .

Solution: $0.5 \times 10^4 = 5 \times 10^3 = 5,000$.

Apply: Build the Fermi Sheet

Open Google Sheets. New sheet. Name it **Olympus Unit 0**. Build the Rubik's cube estimate from the story, then make it yours.

1. Column A: labels. Column B: numbers. Rows 1 to 4: room length, width, height (cm), cube side (cm).
2. B5: `=B1*B2*B3` (room volume). B6: `=B4^3` (cube volume). B7: `=B5/B6`.
3. Change B1 to 800. Watch B7 double. Change B4 to 3 (a mini cube). Watch it explode.
4. Add a row 8: `=B7*10` for the cost at \$10 per cube. Format as currency (the \$ button).
5. Now build a second block for *your* Fermi problem from yesterday, one piece per row, multiplied at the bottom.
6. Optional, five minutes: go to colab.research.google.com, click New Notebook, type `print(2**10)`, press the play button. That's Python. It says 1024. You'll meet it properly in Unit 2.

Dad notes: Watch for the classic beginner move: typing the calculated number into a cell instead of a formula. The sheet should recalculate when an input changes. If it doesn't, it's a poster, not a model.

Week 1 · Day 4 · Lab Day: Lab Day: The First Labor

Labor 0, Quest 1, and the trick goes public

Warm-up

Head math: **15% of 80?**

Answer: 12. 10% of 80 is 8, 5% is half that (4), together 12. That's how servers do tips in their head. Unit 2.

The Story

What a Labor is

Heracles had twelve. You'll have one per unit. A Labor is a multi-part problem that uses everything from the unit, with **no hints**. You can retry. You can use paper, calculator, Sheets. You cannot use Claude. Beat every part and the unit badge is yours.

Labor 0 is on the Labor page (link at the bottom). Do it after this page.

What a Quest is

THE STORY · FEYNMAN, SAFECRACKER

At Los Alamos, during the war, Feynman got bored and taught himself to open the filing cabinets that held the atomic bomb's secrets. Not with tools: with math and psychology. He worked out that the locks had far fewer real combinations than advertised, that people left them set near their combination, and that physicists picked combinations like 27-18-28 (the digits of the number e). He'd open a colleague's safe and leave a note inside. Security was not amused.

Your Quests are the friendly version. Every clue is a problem. The answer tells you where the next clue is hidden. The final answers open a real lock on a real box. Dad hides everything before you start. Quest 1 is short; it's a tutorial. Later ones get mean.

The trick goes public

MATHEMAGIC · WILLMAN BADGE, ATTEMPT 1

Tonight: perform "I can read your mind" for at least two people. Then explain, with the n-table, why it works. Dad awards the badge if the explanation is clear enough that *they* could do it. Sell it. Pause before the reveal. Willman doesn't just say "five"; he makes you wait for it.

ON PAPER

Week 1 review in the notebook: Codex should have at least 8 terms. Write one sentence about the most useful thing you learned this week. It's fine if it's "the trick."

Practice

1. A quick Fermi check before the Labor: the real Millennium Falcon is about **34.75 m** long. The big Lego set (75192) is about **84 cm** long. Roughly how many times longer is the real ship? (One significant figure is fine.)

Answer: 41 (± 2)

Hints: (1) Same units first: 84 cm = 0.84 m. (2) $34.75 \div 0.84$.

Solution: $34.75 \div 0.84 \approx 41$. The set is about 1:41 scale. Hold that number; the Labor needs it.

2. If every *length* gets multiplied by 41, what does the *volume* get multiplied by? (Think of a cube: length $\times$ width $\times$ height, each one 41 $\times$ bigger.)

Answer: 68,921 (± 100)

Hints: (1) Volume has three lengths multiplied together. (2) $41 \times 41 \times 41$.

Solution: $41^3 = 68,921$. Scaling a length by 41 scales volume by 41^3 , about 69,000. This is the single most important idea in Unit 3 (why elephants can't jump), so it's fine to be surprised by it now.

3. Bug Hunter practice. An AI says: "41 cubed is 68,291." Is it right?

- A. Yes
- B. No, it's 68,921 (digits swapped)
- C. No, it's 6,892

Answer: No, it's 68,921 (digits swapped) (B)

Hints: (1) Compute it yourself; you just did.

Solution: 68,921. The AI swapped two digits. That's exactly the kind of small, confident error you'll catch all year.

Apply: Labor 0 and Quest 1

Two things today, in this order:

1. **Labor 0: The Fermi Falcon.** Go to the Labor page (Unit 0 $\rightarrow$ Labor 0). Four parts, no hints. Beat it and the Initiate badge is yours.
2. **Quest 1.** Dad has hidden the clues. Clue 1 is on the Quest page. Pencil and brain only.
3. **Perform the trick.** Two people. Explain it. Willman badge.
4. **Check the cloud chip** at the top of the page says **saved**. Your progress follows you to any computer with your name and PIN. (If the chip says offline, use Progress $\rightarrow$ Save progress file as a backup.)

Dad notes: Quest 1 setup is on the Quest page in Dad mode (answers, hiding spots, lock combo). Print the clue cards from there. Put something real in the box: a treat, a new cube, cash for the

calculator. The lock should be a cheap 3-digit luggage lock or a dial padlock; both work with the numbers given. Award the Willman badge from the Badges page if the explanation lands.

Labor 0: The Fermi Falcon

On Day 2 you estimated the bricks in a life-size Millennium Falcon from its volume. Now check it a completely different way: scale up the real Lego set. If two methods agree, you can trust them both. No hints. Paper, calculator, Sheets allowed.

Part 1. Part 1. The Lego UCS Millennium Falcon (set 75192) is about **84 cm** long. The "real" Falcon is about **34.75 m**. Find the scale factor (how many times longer the real one is). Two decimal places.

Answer: 41.37 (± 0.15)

Solution: $34.75 \div 0.84 = 41.37$.

Part 2. Part 2. If each length is $41.37\times$ longer, volume is 41.37^3 times bigger. Compute that. (Nearest thousand is fine.)

Answer: 70,800 ($\pm 1,500$)

Solution: $41.37^3 \approx 70,800$.

Part 3. Part 3. The set has **7,541 pieces**. If you scaled every piece up, you'd need $7,541 \times 41.37^3$ pieces of *giant* Lego. But you want *normal* bricks, so that's your brick count. Compute it in scientific notation (e.g. $3.2e8$).

Answer: 534,000,000 ($\pm 20,000,000$)

Solution: $7,541 \times 70,800 \approx 5.3 \times 10^8$: about 530 million bricks. Day 2's volume method gave 4 to 5×10^8 . Two different methods, same order of magnitude. That's confirmation.

Part 4. Part 4. Lego averages roughly **10 cents per piece**. What would the bricks cost, in dollars? (Scientific notation or plain number.)

Answer: 53,400,000 dollars ($\pm 3,000,000$)

Solution: $5.3 \times 10^8 \times \$0.10 \approx \5.3×10^7 : about **\$53 million**. For comparison, the actual movie prop cost a lot less. Han Solo won his in a card game, which is the Unit 6 way to acquire a starship.

Quest 1: The Tutorial Quest (Dad setup)

Four clues. Each answer is a number that points to a place in the house. The last three answers open the lock.

Hide the clues in order (clue 2 where clue 1's answer points, and so on). Put the locked box where clue 4's answer points. Set the lock: a 3-digit luggage lock reads **4-2-2** (last digit of clues 1, 2, 3); a dial padlock reads **24-32-12**. Tell Hudson which kind of lock it is. Swap hiding spots for whatever fits your house; the number just needs to suggest the place.

#	Clue	Answer	Hiding spot for the next clue
1	There are 86,400 seconds in a day. Divide by 3,600. Where in the house do you see that number every single day?	24	Taped under or behind the clock (or on the microwave clock).
2	$2^5 = ?$ Water does something important at this many degrees Fahrenheit. Go where that happens.	32	In the freezer, in a zip bag.
3	$12^2 \div 12 = ?$ They come in cartons of exactly this many.	12	In the egg carton in the fridge.
4	$(3 \times 10^2) \div (5 \times 10^1) = ?$ A cube has this many faces; the game version has dots. The box is with the games. The lock wants the last digit of your first three answers (or the full numbers, on a dial).	6	The locked box goes on the game shelf, next to the dice.

Lock: 3-digit luggage lock: **4-2-2** (the last digit of the answers to clues 1, 2, 3). Dial padlock: **24-32-12**.